

## Week 2 — Variables & Math

Name: \_\_\_\_ Date: \_\_\_\_

A **variable** is a labeled box in the computer's memory. = puts a value in the box.

### Today's words

| Word                | What it means                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>variable</b>     | a labeled box that holds one value                                                                                        |
| <b>assignment</b>   | putting a value in a box: <code>score = 10</code>                                                                         |
| <b>reassignment</b> | replacing what's in the box ( <code>score = score + 10</code> — take, add, put back!)                                     |
| <b>expression</b>   | a math phrase Python can work out, like <code>3 * x + 2</code>                                                            |
| <b>operator</b>     | a math symbol: <code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>**</code> <code>//</code> <code>%</code> |

### 1 • Be the computer 🤖

Follow the code *exactly* and write what the screen shows. (Careful on line 3!)

```
x = 10
print(x)
print("x")
x = x + 5
print(x)
```

Line 1: \_ Line 2: \_ Line 3: \_

### 2 • PEMDAS drill — evaluate like Python 🇺🇸

```
print(2 + 3 * 4)      # → _____
print((2 + 3) * 4)    # → _____
print(10 - 2 * 3)     # → _____
print(2 ** 3)         # → _____
```

### 3 • The pizza operators 🍕

23 slices arrive for 4 kids.

```
print(23 // 4)  # slices EACH: _____
print(23 % 4)   # slices LEFT OVER: _____
```

Check yourself: (each × 4) + left over should equal \_\_\_\_.

### 🧠 Brain teaser (optional — take it home)

**The two-box swap.** Real programmers get asked this in job interviews!

Box `a` holds **3**. Box `b` holds **7**. Make them trade values — `a` ends with 7 and `b` ends with 3. Rules:

- Every step must be an assignment (something like `box = box`).
- A box holds only ONE value — putting something in destroys what was there.
- You can't "remember" a number in your head. Boxes only!

Hint: `a = b` first destroys the 3 forever... so maybe you need a third box? Write your steps like code. Bring it next week for a shout-out.